1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist flagged that MIBC tumors with apparent local immune organization (high Immunoscore/TLS or TILs) do not uniformly show favorable outcomes — why does immune presence sometimes fail to control progression? 2) Tier 1 paths used: I built on the TGFB1 → stromal/fibrogenic remodeling path (Core Path 1) because TGF-β can create a physical and suppressive stromal niche, and on the immune-context/Immunoscore axis (Core Path 2) using CD8A as a marker of effector T cells/TLS effectiveness; these directly address how stroma and immune infiltration interact. 3) Tier 2 support: PTGDS (lipid/inflammatory modulation), SDC2/TFPI2 (epigenetic detectability in non-invasive samples), SARIFA (stromal–adipocyte niche) and ABH glycan remodeling all provide plausible secondary mechanisms that could modulate TLS function or offer alternative assay modalities, reinforcing the core stroma–immune interaction. 4) Endpoint rationale: I selected progression-free interval (PFI) because the puzzle concerns early failure/progression despite immune infiltration — PFI best captures clinically meaningful early tumor escape driven by local microenvironmental suppression. 5) Deliberate exclusions: I excluded AI/translation-aspect, non-invasive assay modalities and broader multi-gene panels from the core hypothesis to keep the mechanistic question sharp; those elements are retained as secondary_modifiers for downstream biomarker development rather than being in the core mechanistic claim?

Working Hypothesis

In muscle-invasive bladder cancer, TLS-rich tumors with high TGFB1 expression exhibit impaired CD8A+ T cell effector infiltration and function due to TGF-β–driven stromal remodeling that disrupts TLS architecture and excludes effector T cells. Consequently, TLS-high/TGFB1-high tumors will have shorter progression-free interval than TLS-high/TGFB1-low tumors, with the magnitude of this effect moderated by secondary modifiers PTGDS, SDC2, TFPI2, ABH, and SARIFA that reflect additional immune-regulatory or stromal cues and can serve as biomarkers for stratification or combination therapies.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	TGFB1, CD8A
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1 → stromal/fibrogenic remodeling path (Core Path 1) because TGF-β can create a physical and suppressive stromal niche, and on the immune-context/Immunoscore axis (Core Path 2) using CD8A as a marker of effector T cells/TLS effectiveness; these directly address how stroma and immune infiltration interact.

CD8A: CD8A as a marker of effector T cells/TLS effectiveness; these directly address how stroma and immune infiltration interact.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	ABH_antigens	Supporting modifier from knowledge graph
Secondary	SARIFA_signature	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses in MIBC cohorts to quantify TLS presence/maturation by imaging/pathology and TGFB1/CD8A expression;

Prediction 2: survival analysis comparing TLS-high/TGFB1-high vs TLS-high/TGFB1-low with PFI as endpoint;

Prediction 3: validation in independent datasets;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PTGDS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- SDC2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.